

VAMSHI SHIRAGANI

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PROFESSIONAL SUMMARY

- Java Backend Developer with 5+ years of experience designing, developing, and deploying scalable backend solutions and microservices architectures.
- Expertise in building RESTful APIs, Spring Boot microservices, serverless and cloud-native applications using AWS (Lambda, SQS, ECS, EKS) and Kafka.
- Skilled in SQL/NoSQL database design, query optimization, caching strategies, and performance tuning to ensure high availability and fast API response times.
- Proven ability to implement CI/CD pipelines, automated testing (JUnit, Mockito), and secure authentication mechanisms (OAuth 2.0, JWT) for enterprise-grade applications.
- Experienced in collaborating with Agile teams and integrating backend services with React.js, Redux, and Context API to deliver seamless end-to-end solutions.

TECHNICAL SKILLS

Languages: Java, SQL, PL/SQL, JavaScript, TypeScript, J2EE.

Backend Frameworks & Libraries: Spring Boot, Spring MVC, Spring REST, Spring WebFlux, Spring Data JPA, Spring Security, Hibernate, JUnit, Mockito, GraphQL.

Front-End Technologies: React.js, Redux, Context API, HTML5, CSS3, JavaScript ES6+.

Databases & Data Management: Oracle, PostgreSQL, MongoDB, DynamoDB, MySQL, Database Design, Query Optimization, Indexing.

Messaging & Event Streaming: Apache Kafka, RabbitMQ, JMS.

Cloud & DevOps Tools: AWS (Lambda, S3, SQS, API Gateway, ECS, EKS, CloudWatch, CloudFormation), Docker, Kubernetes, Jenkins, Git, GitHub, Maven, Gradle, CI/CD pipelines.

Testing & API Validation: JUnit, Mockito, Postman, SOAPUI, REST Assured, Integration Testing, Unit Testing.

Microservices & Architecture: Microservices, RESTful APIs, Serverless Architecture, API Gateway, OAuth 2.0, JWT, Load Balancing, Scalability, High Availability.

Methodologies: Agile, Scrum, SDLC, Continuous Integration, Continuous Deployment.

PROFESSIONAL EXPERIENCE

Java Backend Developer

Oct 2024 – Present

PayPal | Remote, MO

- Engineered backend microservices and synchronized them with a React.js frontend using Redux and Context API, boosting UI responsiveness and reducing data lag by 25%.
- Leveraged Spring Boot and Spring REST to implement scalable APIs, shortening development cycles by 20% while ensuring maintainable code for complex payment workflows.
- Secured sensitive endpoints by integrating OAuth 2.0 with JWT token refresh, enabling safe access for thousands of users across concurrent sessions.
- Designed and deployed Kafka-based message queues to handle microservices communication, improving real-time transaction processing and throughput by 30%.
- Deployed containerized services on AWS EKS using Docker, reducing deployment time by 20% and supporting horizontal scaling for high-demand operations.
- Enhanced the CI/CD process with Jenkins and automated testing, increasing deployment reliability by 10% and minimizing post-release defects.
- Tuned backend queries and optimized data access with Spring Data JPA and indexing strategies, improving API response times by 30% under peak load.
- Conducted thorough API testing using Postman and SOAPUI, proactively identifying issues to maintain 100% uptime and ensure seamless payment operations.

Java Developer II

Jun 2024 – Oct 2024

PNC Financial Services | Remote

- Engineered RESTful APIs using Spring Boot and Spring REST, integrating MongoDB and PostgreSQL to reduce data retrieval latency by 25% and accelerate transaction processing.
- Designed GraphQL schemas for microservices, streamlining frontend data requests and cutting redundant API calls by 30% across multiple banking services.
- Developed serverless backend functions with AWS Lambda and API Gateway, ensuring uninterrupted processing for high-concurrency customer transactions.
- Containerized microservices with Docker and deployed them on Kubernetes, achieving a 40% reduction in infrastructure costs while improving application scalability.
- Enhanced API security by implementing Spring Security and JWT authentication, protecting sensitive customer information and meeting enterprise compliance standards.
- Automated backend testing using JUnit and Mockito, increasing system reliability and reducing production bugs by 35% during release cycles.
- Integrated Kafka messaging for microservices communication, enabling near real-time updates and consistent data synchronization across distributed systems.
- Optimized database queries and indexing with Spring Data JPA, improving API response time by 30% and ensuring smooth performance under peak loads.

Associate Application Developer
Fujitsu Consulting India Pvt Ltd | India

Mar 2021 – Dec 2022

- Engineered RESTful APIs using Spring MVC and WebFlux, boosting data throughput by 30% and enabling smooth handling of concurrent high-volume requests.
- Streamlined business logic and DAO layers with Hibernate and Spring IOC, reducing code errors and accelerating feature development across critical modules.
- Built automated unit and integration tests with JUnit, cutting manual testing effort by 40% and significantly improving backend reliability.
- Designed cloud-native solutions using DynamoDB, supporting thousands of concurrent users while maintaining high availability and fast data access.
- Enhanced API performance by optimizing database queries and implementing caching strategies, reducing response times by 25% during peak workloads.
- Configured CI/CD pipelines with Jenkins and Docker, increasing deployment frequency and delivering stable microservices releases across environments.
- Integrated backend services with React.js front-end modules, improving dynamic data rendering and enhancing overall UI responsiveness for end-users.
- Participated in Agile sprints with cross-functional teams, delivering incremental backend features and ensuring alignment with client requirements and project deadlines.

Java Application Developer

Jan 2019 – Mar 2021

Accenture | India

- Developed scalable RESTful APIs using Spring Boot, processing high-volume enterprise requests and increasing system reliability by 30%.
- Implemented Spring MVC controllers with Hibernate DAO layers, improving data consistency and accelerating backend feature delivery by 20%.
- Designed NoSQL cloud solutions using DynamoDB, enabling seamless handling of thousands of concurrent users and improving application uptime.
- Built automated unit and integration tests with JUnit, reducing production defects by 35% and ensuring stability for critical backend services.
- Configured CI/CD pipelines with Jenkins and Docker, improving deployment frequency and maintaining stable releases for microservices.
- Integrated backend services with React.js modules, enhancing dynamic content rendering and improving end-user experience across applications.
- Applied Spring Security and OAuth 2.0 for secure authentication and authorization, safeguarding sensitive enterprise data across services.
- Optimized database queries and RESTful endpoints, reducing response times by 20% and supporting high-concurrency transaction workloads.

PROJECTS

Enterprise Payment Processing Platform

- Developed microservices using Spring Boot and Kafka, processing thousands of payment transactions in real-time with high throughput and reliability.
- Designed secure RESTful APIs integrated with React.js frontend, ensuring seamless data flow and reducing API call errors by 25%.
- Deployed containerized services on AWS EKS with Docker, improving deployment efficiency and enabling horizontal scaling for high-volume traffic.

Cloud-Based Customer Data Management System

- Built backend services using Spring Boot and GraphQL, enabling optimized data retrieval and reducing redundant API requests by 30%.
- Implemented AWS Lambda functions with API Gateway, supporting thousands of concurrent requests with serverless architecture and zero downtime.
- Optimized DynamoDB schema design and automated unit testing, ensuring scalable, cost-efficient cloud storage and reliable data transactions.

Banking Transaction Analytics

- Developed real-time analytics pipeline with Kafka streaming banking transactions, improving data processing speed and reliability.
- Created backend APIs with Spring Boot and Spring Data JPA, enabling secure and efficient access to transactional data.
- Integrated Jenkins CI/CD pipeline and optimized PostgreSQL queries, reducing deployment errors by 30% and improving query response times.

EDUCATION

Masters in Computer and Information Science

Northwest Missouri State University | Maryville, MO

CERTIFICATIONS

- IBM Java Developer Professional Certificate – **Coursera**
- Java Essential Training – **LinkedIn Learning**
- Java Programming: Solving Problems with Software – **Coursera**
- Java Programming and Software Engineering Fundamentals – **Coursera**
- Spring Framework: Spring Boot and Spring Cloud – **LinkedIn Learning**